

Literature Status for arXiv:2308.05671: Arbitrary Amplifications of A^{*n}

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Status

This is a `literature_already_answered` packet, not a new theorem of this run. Dykema and Zhao [2] explicitly answer the two amplification questions in Section 4.3 of Boutonnet–Drimbe–Ioana–Popa [1].

The source questions

Let A be a diffuse non-separable abelian tracial von Neumann algebra. The source proves that the free powers A^{*n} , $2 \leq n \leq \infty$, are mutually non-isomorphic, and that finite free powers have trivial fundamental group. In Section 4.3 it asks for an identification of $(A^{*n})^t$ at arbitrary $t > 0$. For homogeneous A it proves only that, for $n \geq 2$ and $k \in \mathbb{N}$,

$$(A^{*n})^{1/k} \cong A^{*(nk)} * LF_{1+nk(k-1)-k^2}.$$

It also obtains $\mathbb{Q} \subseteq \mathcal{F}(A^{*\infty})$ and records the expectation $\mathcal{F}(A^{*\infty}) = \mathbb{R}_+^*$.

The later all-scale answer

Dykema–Zhao first prove that every diffuse abelian tracial von Neumann algebra is *self-symmetric* (Theorem 2.8 of [2]); thus their hypothesis applies to every A in the source problem, without an additional homogeneity assumption. They then define factors $\mathcal{F}_{s,r}(A)$ for

$$s > 0, \quad 1 - s < r \leq \infty,$$

such that $\mathcal{F}_{n,0}(A) \cong A^{*n}$ for every integer $n \geq 2$. Their Theorem 3.5 proves the rescaling formula

$$(\mathcal{F}_{s,r}(A))^t \cong \mathcal{F}_{s/t, (s+r-1)/t^2-s/t+1}(A) \quad (t > 0). \quad (1)$$

Specializing (1) to $s = n$ and $r = 0$ gives the requested full identification

$$\boxed{(A^{*n})^t \cong \mathcal{F}_{n/t, (n-1)/t^2-n/t+1}(A)} \quad (n \geq 2, t > 0). \quad (2)$$

For $t = 1/k$, the first parameter becomes the integer nk , and (2) recovers the source formula.

Finally, Theorem 4.5 of [2] proves

$$\boxed{\mathcal{F}(A^{*\infty}) = \mathbb{R}_+^*},$$

upgrading the source’s rational subgroup to the full positive multiplicative group. The abstract of [2] explicitly says that these conclusions answer two questions in Section 4.3 of [1].

Why the interpolation is essential

The sans-rank amplification law assigns $(A^{*n})^t$ non-separable rank n/t . When n/t is not an integer, no formula involving only an ordinary integer number of unital copies of A can encode that parameter. The family $\mathcal{F}_{s,r}(A)$ supplies exactly the missing continuous non-separable multiplicity and proves that it behaves coherently under every rescaling.

Scope

The amplification subsection is fully resolved. This status packet makes no claim about the source's distinct questions concerning freely complemented maximal amenable MASAs, the outer automorphism group of A^{*n} , or finite-index subfactors.

References

- [1] R. Boutonnet, D. Drimbe, A. Ioana, and S. Popa, *Non-isomorphism of A^{*n} , $2 \leq n \leq \infty$, for a non-separable abelian von Neumann algebra A* , *Geom. Funct. Anal.* 34 (2024), 393–408; arXiv:2308.05671.
- [2] K. Dykema and J. Zhao, *Free products and rescalings involving non-separable abelian von Neumann algebras*, *J. Funct. Anal.* 290 (2026), article 111264; arXiv:2502.08595; doi:10.1016/j.jfa.2025.111264.